The disc that the seed's own angular momentum would make

A companion note to *From a primordial fluctuation to an idealised rotation curve*

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Abstract

In the seed of the main paper every shell rotates rigidly with the specific angular momentum it receives by turnaround, $j \propto M$. If the gas of each mass element keeps its own angular momentum and settles on a circular orbit in the final rotation curve, the disc that forms is fixed completely by the seed. We derive its surface density in closed form for a flat rotation curve, compute it numerically for the P18 curve with its core, and compare it with an exponential disc and with the relation of Mo et al. (1998) used in the main paper. The seed disc is close to exponential between one and four scale lengths, but it has twice the exponential mass inside half a scale length and a sharp edge at $R_{\max} = 3\sqrt{2}\lambda GM/v \simeq 4.3\,R_d$. Its scale length is 1.06 to 1.4 times the Mo–Mao–White value, depending on how R_d is defined for a non-exponential disc, and it depends only weakly on the spin. None of this changes the conclusions of the main paper; it lowers the inferred retained fraction of angular momentum f_j and leaves the failed spin test as it is.

1 Motivation

The main paper tests the energy closure with discs: if the stellar disc keeps a fraction f_j of the seed's specific angular momentum, then $R_d/r_c = f_j\lambda y/\sqrt{2}$ (its Sect. 6.5). This relation assumes an exponential disc in a flat rotation curve, for which the specific angular momentum is $j_d = 2vR_d$ (Fall & Efstathiou, 1980; Mo et al., 1998). But the seed carries more information than its mean angular momentum: it fixes how the angular momentum is *distributed* over its mass. Under detailed conservation, each mass element keeping its own j (Mestel, 1963; Fall & Efstathiou, 1980), this distribution fixes the disc. The question of this note is whether the disc so obtained is exponential, and how its scale length compares with the Mo–Mao–White value.

2 Theory

2.1 Angular momentum of the seed

As in the main paper we use units with $G = v = 1$, where v is the flat speed, and measure masses in units of the seed mass $M_{\rm tot}$. A shell enclosing the mass M turns around at $R_{\rm ta} = 2GM/v^2$ and receives by then, through tidal torques, the specific angular momentum

$$j_{\rm sh}(M) = \sqrt{2}\,\lambda\,vR_{\rm ta} = 2\sqrt{2}\,\lambda\,\frac{GM}{v}, \tag{1}$$

chosen so that the total angular momentum, $J = \int_0^M j_{\rm sh}\,{\rm d}M' = \sqrt{2}\lambda GM^2/v$, returns the spin parameter of Peebles (1969), $\lambda = J|E|^{1/2}/(GM^{5/2})$ with $|E| = \frac{1}{2}v^2M$, for every M. Each shell rotates rigidly about a common axis z. A mass element at polar angle θ has $j_z = \omega R^2\sin^2\theta$, and averaging $\sin^2\theta$ over the sphere ($\langle\sin^2\theta\rangle = \frac{2}{3}$) gives $j_{\rm sh} = \frac{2}{3}\omega R^2$; hence

$$j_z = \tfrac{3}{2}\,j_{\rm sh}(M)\,\sin^2\theta = 3\sqrt{2}\,\lambda\,\frac{GM}{v}\,(1-\mu^2), \qquad \mu = \cos\theta. \tag{2}$$

Since the shells are equally spaced in M and the mass of a shell is spread uniformly in μ, the seed is a population of mass elements with $M/M_{\rm tot}$ and μ uniformly distributed in $[0,1]$. The largest angular momentum, on the equator of the outermost shell, is $j_{\rm max} = 3\sqrt{2}\lambda G M_{\rm tot}/v$, and the mean is $\langle j_z\rangle = j_{\rm max}\langle M/M_{\rm tot}\rangle\langle 1-\mu^2\rangle = j_{\rm max}\cdot\frac{1}{2}\cdot\frac{2}{3} = \sqrt{2}\lambda G M_{\rm tot}/v$, the specific angular momentum of the seed.

2.2 Detailed conservation and the disc

Let a fraction of the matter be gas that keeps a fraction f_j of its own j_z and settles on the circular orbit with that angular momentum in the final rotation curve $v_c(R)$:

$$R\,v_c(R) = f_j\,j_z. \tag{3}$$

Then the distribution of j_z over the mass becomes the distribution of disc mass over radius. We take the gas to be a fixed fraction of every mass element, so the disc mass M_d is distributed as the seed's j_z.

Flat rotation curve (analytic). With $v_c = v$ the orbit radius is $R = f_j j_z/v$, and with $S = R/R_{\rm max}$,

$$R_{\rm max} = \frac{f_j j_{\rm max}}{v} = 3\sqrt{2}\,f_j\lambda\,\frac{GM_{\rm tot}}{v^2}, \qquad S = \frac{M}{M_{\rm tot}}\,(1-\mu^2). \tag{4}$$

The disc mass inside S is the fraction of the unit square $(M/M_{\rm tot},\mu)$ where $(M/M_{\rm tot})(1-\mu^2) < S$. For $1-\mu^2 < S$, i.e. $\mu > \sqrt{1-S}$, every M qualifies; otherwise only $M/M_{\rm tot} < S/(1-\mu^2)$. Hence

$$F(S) \equiv \frac{M_d(<R)}{M_d} = \left(1-\sqrt{1-S}\right) + \int_0^{\sqrt{1-S}} \frac{S\,{\rm d}\mu}{1-\mu^2} = 1-\sqrt{1-S} + S\ {\rm artanh}\,\sqrt{1-S}, \tag{5}$$

using $\int {\rm d}\mu/(1-\mu^2) = {\rm artanh}\,\mu$. Differentiating, the two terms $\pm 1/(2\sqrt{1-S})$ cancel and $F'(S) = {\rm artanh}\,\sqrt{1-S}$, so the surface density, $\Sigma = ({\rm d}M_d/{\rm d}R)/(2\pi R)$, is

$$\Sigma(R) = \frac{M_d}{2\pi R_{\rm max}^2}\,\frac{{\rm artanh}\,\sqrt{1-S}}{S}, \qquad S = \frac{R}{R_{\rm max}} \le 1. \tag{6}$$

Two limits show its character. Near the centre, ${\rm artanh}\,\sqrt{1-S} \simeq \frac{1}{2}\ln(4/S)$, so $\Sigma \propto \ln(4/S)/S$: a central cusp, steeper than any exponential. At the edge, ${\rm artanh}\,\sqrt{1-S} \simeq \sqrt{1-S}$, so $\Sigma \to 0$ as $\sqrt{1-S}$: the disc ends sharply at $R_{\rm max}$, the orbit of the equator of the outermost shell. In between, Eq. (6) is close to exponential (Fig. 1a).

P18 rotation curve (numerical). Inside the core the curve rises, $v_c \simeq (2/\sqrt{3})(v/r_c)R$, so Eq. (3) gives $R \propto j_z^{1/2}$ instead of $R \propto j_z$: low-j material lands further out than in a flat curve, and the central cusp of Eq. (6) is softened. We invert Eq. (3) on a grid for the P18 curve $v_c = v\sqrt{m(x)/x}$, $x = R/r_c$, with $r_c = R_b/y$ and the core ratio y of the main paper (its Eq. 17, $\Lambda = 0$; $y = 50.5$ for $\lambda = 0.035$), and sample 2×10^6 mass elements from the uniform distribution of $(M/M_{\rm tot},\mu)$.

Scale length of a non-exponential disc. For the comparison with an exponential disc a scale length must be defined. We use two definitions: an exponential fit to Σ between one and four scale lengths (iterated), as photometry would measure it, and the half-mass radius divided by 1.678, its value for an exponential disc. The relation of Mo et al. (1998) for a flat curve gives $R_d^{\rm MMW} = f_j\langle j_z\rangle/(2v) = f_j\lambda G M_{\rm tot}/(\sqrt{2}v^2)$, i.e. $R_d/r_c = f_j\lambda y/\sqrt{2}$ with $GM_{\rm tot}/v^2 \simeq R_b$.

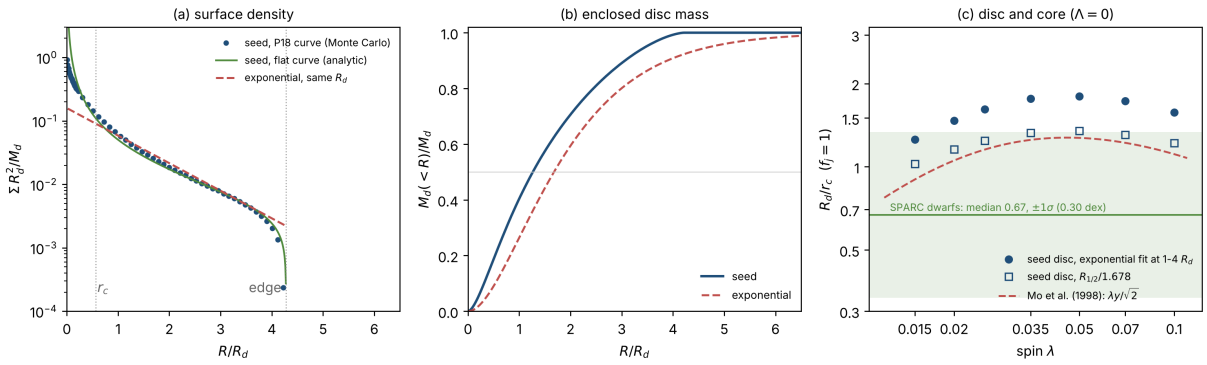


Figure 1: The disc made by the seed's own angular momentum ($f_j = 1$, $\Lambda = 0$). (a) Surface density for $\lambda = 0.035$: Monte Carlo in the P18 curve (points), the analytic flat-curve result, Eq. (6) (solid), and an exponential disc with the same scale length (dashed); dotted lines mark the core radius and the edge $R_{\max}$. (b) Enclosed disc mass, seed against exponential. (c) Ratio of disc scale length to core radius against spin: seed disc with R_d from an exponential fit at 1–4 R_d (filled) and from the half-mass radius (open), and the relation of Mo et al. (1998) used in the main paper (dashed). Green: the median and 1σ spread of the 79 SPARC dwarfs; the observed values require $f_j < 1$.

3 The code

The script `seed_disc.py` (Appendix A) does the following. (i) It solves the core equation of the main paper for y. (ii) It draws mass elements with uniform $M/M_{\rm tot}$ and μ, assigns j_z from Eq. (2), and places each on the circular orbit of Eq. (3) in the P18 curve, by interpolating the monotonic function $Rv_c(R)$. (iii) It computes Σ, the enclosed mass and the two scale lengths, and compares them with the analytic flat-curve result, Eqs. (5) and (6), and with an exponential disc. (iv) It repeats the calculation for spins from 0.015 to 0.1 and makes Fig. 1. Near the centre the closed form of $m(x)$ loses precision through cancellation; the code uses the series $m \simeq \frac{4}{3}x^3$ for $x < 10^{-3}$. It runs in about twenty seconds.

4 Results

For the median spin, $\lambda = 0.035$ ($y = 50.5$, $r_c = 0.020\,R_b$), the script gives the following.

- *Shape.* Between one and four scale lengths the disc is close to exponential (Fig. 1a). Inside $0.5\,R_d$ it holds 19% of its mass, twice the 9% of an exponential disc; the half-mass radius is $1.27\,R_d$ instead of $1.68\,R_d$ (Fig. 1b). The disc ends at $R_{\max} = 0.149\,R_b = 4.3\,R_d$.

- *Effect of the core.* The rising inner curve moves low-j material outward: inside $S = 0.1$ the P18 curve places 15% of the disc mass, the flat curve 23%. Beyond $S \simeq 0.3$ the two agree.

- *Scale length.* The exponential fit gives $R_d = 0.035\,R_b$, $R_d/r_c = 1.76 f_j$, 1.40 times the Mo–Mao–White value $1.26 f_j$; the half-mass definition gives $1.33 f_j$, 1.06 times that value. The difference between the two definitions measures how far the seed disc is from exponential.

- *Dependence on spin.* R_d/r_c changes little with λ (Fig. 1c): $\mathrm{d}\ln(R_d/r_c)/\mathrm{d}\ln\lambda = 0.16$ at $\lambda = 0.035$, with a broad maximum near $\lambda \simeq 0.05$. The reason is the same as in the main paper: a larger spin makes the disc larger but also the core larger (y smaller), and the two nearly cancel.

5 Relevance to the main paper

The retained fraction f_j. The observed median $R_d/r_c = 0.67$ is reproduced with a smaller f_j than with the Mo–Mao–White relation: smaller by the factor 1.06 to 1.4, so the $f_j \simeq 0.77$ of the main paper becomes 0.55–0.73. This is within the range usually inferred for discs and does not change any conclusion.

The spin test. The test of the main paper rests on the *width* of the distribution of R_d/r_c. A constant factor does not change the width, and the weak dependence on λ found here (0.16) is the same as with the Mo–Mao–White relation. The predicted distribution therefore stays as narrow as before, and the spin test fails as before.

The shape of real discs. Detailed conservation of the seed's angular momentum gives too much mass at the centre and a sharp edge, the well-known angular-momentum distribution problem of discs that form by detailed conservation (Dalcanton et al., 1997; Bullock et al., 2001; van den Bosch et al., 2001). Late-type dwarfs are bulgeless, so the central excess must have been removed or redistributed; preferential removal of low-angular-momentum gas by feedback is the usual explanation (Governato et al., 2010; Brook et al., 2011). The edge near $4.3\,R_d$ is reminiscent of the truncations and breaks observed in disc profiles at three to five scale lengths (van der Kruit, 1979; Pohlen & Trujillo, 2006), but a dissipational disc need not keep the sharp edge of Eq. (6).

References

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A Code

```
"""The disc that the seed's own angular momentum would make.

Seed of pf2irc: shells of mass M with one specific energy -v^2/2 and spin lambda.
Each shell rotates rigidly with the specific angular momentum it has at turnaround,
    j_shell(M) = 2 sqrt2 lambda G M / v ,
so a mass element at polar angle theta has j_z = (3/2) j_shell sin^2(theta).
Detailed angular-momentum conservation: the gas element settles on the circular orbit
    R v_c(R) = f_j j_z
```

```
of the P18 rotation curve v_c = v sqrt(m(x)/x), x = R/r_c, with r_c = R_b / y.

Units: G = v = 1, total seed mass M_tot = 1, so R_b(1 + 1/4y) = 1.
Outputs: numbers printed, figure fig_seed_disc.png.
Run: python3 seed_disc.py (numpy, scipy, matplotlib; ~20 s)
"""
import numpy as np
import matplotlib.pyplot as plt
from scipy.special import lambertw

CE = -0.91737
def y_of(lam, ell=0.0): # core ratio of pf2irc, Eq. (lambert)
    A = 3*lam**2 + 2*ell/3; B = 0.25 - CE - 0.75*lam**2
    return lambertw(4*A*np.exp(4*B)).real/(4*A)
def m(x):
    x = np.asarray(x, float)
    full = x*(1-np.exp(-x))**2 + 0.25*(1-(1+2*x+2*x*x)*np.exp(-2*x))
    return np.where(x < 1e-3, 4*x**3/3, full) # series near the centre avoids round-off
def vc(R, rc): return np.sqrt(m(R/rc)*rc/R) # v_c / v

def F_flat(S): # disc mass fraction inside S = R/R_max for a flat curve (analytic)
    S = np.clip(S, 1e-12, 1); u = np.sqrt(1-S)
    return 1 - u + S*np.arctanh(np.minimum(u, 1-1e-15))
def Sigma_flat(S): # surface density in units of M_d / R_max^2 (analytic)
    S = np.clip(S, 1e-12, 1-1e-15); return np.arctanh(np.sqrt(1-S))/(2*np.pi*S)

def disc(lam=0.035, y=None, ell=0.0, fj=1.0, N=2_000_000, seed=1):
    """Monte Carlo disc radii (units of R_b) for the P18 curve; returns R, R_b, r_c, R_max."""
    y = y_of(lam, ell) if y is None else y
    Rb = 1/(1+1/(4*y)); rc = Rb/y
    rng = np.random.default_rng(seed)
    M = rng.uniform(0, 1, N); mu = rng.uniform(-1, 1, N)
    jz = fj*1.5*2*np.sqrt(2)*lam*M*(1-mu**2)
    Rg = np.logspace(-7, 1, 6000); Jg = Rg*vc(Rg, rc) # invert R v_c(R) = j_z
    R = np.interp(jz, Jg, Rg)
    Rmax = fj*3*np.sqrt(2)*lam # j_max / v
    return R/Rb, Rb, rc/Rb, Rmax/Rb, y

def exp_fit(R, lo=1.0, hi=4.0, Rd0=None):
    """Exponential scale length from the surface density between lo and hi times R_d (iterated)."""
    Rd = np.mean(R)/2 if Rd0 is None else Rd0
    for _ in range(20):
        e = np.linspace(lo*Rd, hi*Rd, 25); h, _ = np.histogram(R, e)
        S = h/(np.pi*(e[1:]**2-e[:-1]**2)); Rm = 0.5*(e[1:]+e[:-1]); ok = S > 0
        Rd = -1/np.polyfit(Rm[ok], np.log(S[ok]), 1)[0]
    return Rd

if __name__ == "__main__":
    lam = 0.035
    R, Rb, rc, Rmax, y = disc(lam)
    Rd = exp_fit(R); Rd_half = np.median(R)/1.678 # two definitions of R_d for a non-exponential disc
    jmean = np.sqrt(2)*lam # <j_z> = sqrt2 lam G M_tot / v (per M_tot)
    Rd_mmw = jmean/2*(1+1/(4*y)) # exponential disc in a flat curve: j = 2 v R_d (units of R_b)
    print(f"lambda = {lam}, y = {y:.1f}, r_c = {rc:.4f} R_b, R_max = {Rmax:.4f} R_b")
    print(f"exponential fit (1-4 R_d): R_d = {Rd:.4f} R_b = {Rd/Rmax:.3f} R_max; R_d/r_c = {Rd/rc:.2f}")
    print(f"Mo-Mao-White (j_d = 2 v R_d): R_d = {Rd_mmw:.4f} R_b; R_d/r_c = {Rd_mmw/rc:.2f}; ratio {Rd/
        Rd_mmw:.3f}")
    print(f"half-mass radius: {np.median(R)/Rd:.3f} R_d (exponential 1.678); edge R_max = {Rmax/Rd:.2f} R_d"
        )
    print(f"R_d from the half-mass radius (R_half/1.678) = {Rd_half:.4f} R_b; R_d/r_c = {Rd_half/rc:.2f};
        ratio to MMW {Rd_half/Rd_mmw:.3f}")
    e = np.array([0, 0.1, 0.25, 0.5])*Rd
    inner = np.histogram(R, e)[0]/len(R)
    from math import exp
    expo = [ (1-(1+b)*np.exp(-b)) - (1-(1+a)*np.exp(-a)) for a, b in zip(e[:-1]/Rd, e[1:]/Rd)]
    print("mass in R/R_d = 0-0.1, 0.1-0.25, 0.25-0.5: seed", inner.round(4), " exponential", np.round(expo,
        4))
    # flat-curve analytic check
    S = np.sort(R)/Rmax; Fmc = np.arange(1, len(S)+1)/len(S); idx = np.searchsorted(S, [0.1, 0.3, 0.6])
    print("analytic flat-curve F(S) at S=0.1,0.3,0.6:", F_flat(np.array([0.1, 0.3, 0.6])).round(4),
        " Monte Carlo (P18 curve):", Fmc[idx].round(4))
    # spin dependence of R_d/r_c
    lams = np.array([0.015, 0.02, 0.025, 0.035, 0.05, 0.07, 0.1])
    q_seed, q_half, q_mmw = [], [], []
```

```
    for l_ in lams:
        R_, Rb_, rc_, Rm_, y_ = disc(l_, N=600_000)
        q_seed.append(exp_fit(R_)/rc_); q_half.append(np.median(R_)/1.678/rc_)
        q_mmw.append(np.sqrt(2)*l_/2*(1+1/(4*y_))/rc_)
    q_seed, q_half, q_mmw = np.array(q_seed), np.array(q_half), np.array(q_mmw)
    print("R_d/r_c (seed fit 1-4 R_d / seed half-mass / MMW):",
          " ".join(f"{a}:{b:.2f}/{d:.2f}/{c:.2f}" for a, b, d, c in zip(lams, q_seed, q_half, q_mmw)))
    print("d ln(R_d/r_c)/d ln lambda (seed, around 0.035):",
          np.round(np.gradient(np.log(q_seed), np.log(lams))[3], 3))

    # ---------------------------------------------------------------- figure
    C1, C2, C3 = "#1f4e79", "#c0504d", "#4f8a3c"
    fig, ax = plt.subplots(1, 3, figsize=(12.5, 3.9))
    # (a) surface density
    edges = np.concatenate([np.linspace(0, 0.2, 21), np.linspace(0.25, 6.5, 60)])*Rd
    h, _ = np.histogram(R, edges); Rm = 0.5*(edges[1:]+edges[:-1])
    Sig = h/len(R)/(np.pi*(edges[1:]**2-edges[:-1]**2))
    x = Rm/Rd; ok = Sig > 0
    ax[0].semilogy(x[ok], Sig[ok]*Rd**2, "o", ms=3, c=C1, label="seed, P18 curve (Monte Carlo)")
    xs = np.linspace(0.02, Rmax/Rd*0.999, 400)
    ax[0].semilogy(xs, Sigma_flat(xs*Rd/Rmax)*(Rd/Rmax)**2, c=C3, lw=1.3, label="seed, flat curve (analytic)
        ")
    ax[0].semilogy(xs, np.exp(-xs)/(2*np.pi), c=C2, ls="--", lw=1.5, label=r"exponential, same $R_d$")
    ax[0].axvline(rc/Rd, c="0.6", lw=0.8, ls=":"); ax[0].text(rc/Rd*1.05, 2e-4, r"$r_c$", color="0.4")
    ax[0].axvline(Rmax/Rd, c="0.6", lw=0.8, ls=":"); ax[0].text(Rmax/Rd*0.83, 2e-4, r"edge", color="0.4")
    ax[0].set(xlabel=r"$R/R_d$", ylabel=r"$\Sigma\,R_d^2/M_d$", xlim=(0, 6.5), ylim=(1e-4, 3))
    ax[0].legend(frameon=False, fontsize=7.5, loc="upper right"); ax[0].set_title("(a) surface density",
        fontsize=10)
    # (b) cumulative mass
    xx = np.linspace(0, 6.5, 300); Fs = np.searchsorted(np.sort(R), xx*Rd)/len(R)
    ax[1].plot(xx, Fs, c=C1, lw=1.8, label="seed")
    ax[1].plot(xx, 1-(1+xx)*np.exp(-xx), c=C2, ls="--", lw=1.5, label="exponential")
    ax[1].axhline(0.5, c="0.8", lw=0.6)
    ax[1].set(xlabel=r"$R/R_d$", ylabel=r"$M_d(<R)/M_d$", xlim=(0, 6.5), ylim=(0, 1.02))
    ax[1].legend(frameon=False, fontsize=8, loc="lower right"); ax[1].set_title("(b) enclosed disc mass",
        fontsize=10)
    # (c) R_d / r_c against lambda
    ll = np.linspace(0.012, 0.11, 200); yy = np.array([y_of(l_) for l_ in ll])
    ax[2].loglog(lams, q_seed, "o", c=C1, ms=5, label=r"seed disc, exponential fit at 1-4 $R_d$")
    ax[2].loglog(lams, q_half, "s", c=C1, mfc="none", ms=5, label=r"seed disc, $R_{1/2}/1.678$")
    ax[2].loglog(ll, np.sqrt(2)*ll/2*(1+1/(4*yy))*yy/(1+1/(4*yy)), c=C2, ls="--", lw=1.4,
                 label=r"Mo et al. (1998): $\lambda y/\sqrt{2}$")
    ax[2].axhline(0.67, c=C3, lw=1.2); ax[2].text(0.0125, 0.69, r"SPARC dwarfs: median 0.67, $\pm1\sigma$
        (0.30 dex)", color=C3, fontsize=7.5)
    ax[2].axhspan(0.67/10**0.30, 0.67*10**0.30, color=C3, alpha=0.12, lw=0)
    from matplotlib.ticker import FixedLocator, NullLocator, FormatStrFormatter
    ax[2].xaxis.set_major_locator(FixedLocator([0.015, 0.02, 0.035, 0.05, 0.07, 0.1])); ax[2].xaxis.
        set_minor_locator(NullLocator())
    ax[2].xaxis.set_major_formatter(FormatStrFormatter("%g"))
    ax[2].yaxis.set_major_locator(FixedLocator([0.3, 0.5, 0.7, 1, 1.5, 2, 3])); ax[2].yaxis.
        set_minor_locator(NullLocator())
    ax[2].yaxis.set_major_formatter(FormatStrFormatter("%g")); ax[2].set_ylim(0.3, 3.2)
    ax[2].set(xlabel=r"spin $\lambda$", ylabel=r"$R_d/r_c$ ($f_j=1$)")
    ax[2].legend(frameon=False, fontsize=7.5, loc="lower right"); ax[2].set_title(r"(c) disc and core ($\
        Lambda=0$)", fontsize=10)
    fig.tight_layout(); fig.savefig("fig_seed_disc.png", dpi=250)
    print("figure written: fig_seed_disc.png")
```

Output (`code/seed_disc_output.txt`):

```
lambda = 0.035, y = 50.5, r_c = 0.0198 R_b, R_max = 0.1492 R_b
exponential fit (1-4 R_d): R_d = 0.0349 R_b = 0.234 R_max; R_d/r_c = 1.76
Mo-Mao-White (j_d = 2 v R_d): R_d = 0.0249 R_b; R_d/r_c = 1.26; ratio 1.402
half-mass radius: 1.265 R_d (exponential 1.678); edge R_max = 4.28 R_d
R_d from the half-mass radius (R_half/1.678) = 0.0263 R_b; R_d/r_c = 1.33; ratio to MMW 1.057
mass in R/R_d = 0-0.1, 0.1-0.25, 0.25-0.5: seed [0.017 0.055 0.1148] exponential [0.0047 0.0218 0.0637]
analytic flat-curve F(S) at S=0.1,0.3,0.6: [0.2332 0.5263 0.8148] Monte Carlo (P18 curve): [0.1529 0.5062
    0.822 ]
R_d/r_c (seed fit 1-4 R_d / seed half-mass / MMW): 0.015:1.25/1.02/0.90 0.02:1.47/1.16/1.06
    0.025:1.62/1.24/1.16 0.035:1.76/1.33/1.26 0.05:1.80/1.35/1.28 0.07:1.73/1.31/1.23 0.1:1.57/1.22/1.13
d ln(R_d/r_c)/d ln lambda (seed, around 0.035): 0.16
figure written: fig_seed_disc.png
```